

Running

```
# gdb <program> [core dump]
    Start GDB (with optional core dump).

# gdb --args <program> <args...>
    Start GDB and pass arguments

# gdb --pid <pid>
    Start GDB and attach to process.

set args <args...>
    Set arguments to pass to program to
    be debugged.

run
    Run the program to be debugged.

kill
    Kill the running program.
```

Breakpoints

```
break <where>
    Set a new breakpoint.

delete <breakpoint#>
    Remove a breakpoint.

clear
    Delete all breakpoints.

enable <breakpoint#>
    Enable a disabled breakpoint.

disable <breakpoint#>
    Disable a breakpoint.
```

Watchpoints

```
watch <where>
    Set a new watchpoint.

delete/enable/disable <watchpoint#>
    Like breakpoints.
```

<where>

```
function_name
    Break/watch the named function.

line_number
    Break/watch the line number in the cur-
    rent source file.

file:line_number
    Break/watch the line number in the
    named source file.
```

Conditions

```
break/watch <where> if <condition>
    Break/watch at the given location if the
    condition is met.
    Conditions may be almost any C ex-
    pression that evaluate to true or false.

condition <breakpoint#> <condition>
    Set/change the condition of an existing
    break- or watchpoint.
```

Examining the stack

```
backtrace
where
    Show call stack.

backtrace full
where full
    Show call stack, also print the local va-
    riables in each frame.

frame <frame#>
    Select the stack frame to operate on.
```

Stepping

```
step
    Go to next instruction (source line), di-
    ving into function.
```

```
next
    Go to next instruction (source line) but
    don't dive into functions.

finish
    Continue until the current function re-
    turns.

continue
    Continue normal execution.
```

Variables and memory

```
print/format <what>
    Print content of variable/memory locati-
    on/register.

display/format <what>
    Like „print“, but print the information
    after each stepping instruction.

undisplay <display#>
    Remove the „display“ with the given
    number.

enable display <display#>
disable display <display#>
    En- or disable the „display“ with the gi-
    ven number.

x/nfu <address>
    Print memory.
    n: How many units to print (default 1).
    f: Format character (like „print“).
    u: Unit.

    Unit is one of:
        b: Byte,
        h: Half-word (two bytes)
        w: Word (four bytes)
        g: Giant word (eight bytes)).
```

Format

<i>a</i>	Pointer.
<i>c</i>	Read as integer, print as character.
<i>d</i>	Integer, signed decimal.
<i>f</i>	Floating point number.
<i>o</i>	Integer, print as octal.
<i>s</i>	Try to treat as C string.
<i>t</i>	Integer, print as binary (<i>t</i> = „two“).
<i>u</i>	Integer, unsigned decimal.
<i>x</i>	Integer, print as hexadecimal.

<what>

expression

Almost any C expression, including function calls (must be prefixed with a cast to tell GDB the return value type).

file_name::variable_name

Content of the variable defined in the named file (static variables).

function::variable_name

Content of the variable defined in the named function (if on the stack).

{type}address

Content at *address*, interpreted as being of the C type *type*.

\$register

Content of named register. Interesting registers are \$esp (stack pointer), \$ebp (frame pointer) and \$eip (instruction pointer).

Threads

thread <thread#>

Chose thread to operate on.

Manipulating the program

set var <variable_name>=<value>
Change the content of a variable to the given value.

return <expression>
Force the current function to return immediately, passing the given value.

Sources

directory <directory>
Add *directory* to the list of directories that is searched for sources.

list
list <filename>:<function>
list <filename>:<line_number>
list <first>,<last>
Shows the current or given source context. The *filename* may be omitted. If *last* is omitted the context starting at *start* is printed instead of centered around it.

set listsize <count>
Set how many lines to show in „list“.

Signals

handle <signal> <options>
Set how to handle signles. Options are:
(no)print: (Don't) print a message when signals occurs.
(no)stop: (Don't) stop the program when signals occurs.
(no)pass: (Don't) pass the signal to the program.

Informations

disassemble
disassemble <where>
Disassemble the current function or given location.

info args
Print the arguments to the function of the current stack frame.

info breakpoints
Print informations about the break- and watchpoints.

info display
Print informations about the „displays“.

info locals
Print the local variables in the currently selected stack frame.

info sharedlibrary
List loaded shared libraries.

info signals
List all signals and how they are currently handled.

info threads
List all threads.

show directories
Print all directories in which GDB searches for source files.

show listsize
Print how many are shown in the „list“ command.

what is variable_name
Print type of named variable.